Autonomous ECG Cardiac Arrhythmia Classifier

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1. PROJECT OVERVIEW & SCOPE

Comprehensive validation of deep residual 1D-CNN combined with Bidirectional LSTM units on the MIT-BIH Arrhythmia benchmark. Preprocessing included 0.5-40Hz Butterworth bandpass filtering, Pan-Tompkins QRS wave isolation, and data augmentation for ectopic complexes.

2. EXPERIMENTAL BENCHMARKS

Diagnostic accuracy achieved 98.4% across 109,492 cardiac cycles with 97.8% precision, 98.9% recall, and 98.3% F1-score across 5 distinct clinical categories.

3. DEPLOYMENT & VERIFICATION

Inference latency was benchmarked at under 14ms per 10-second segment, suitable for real-time edge telemetry monitoring.